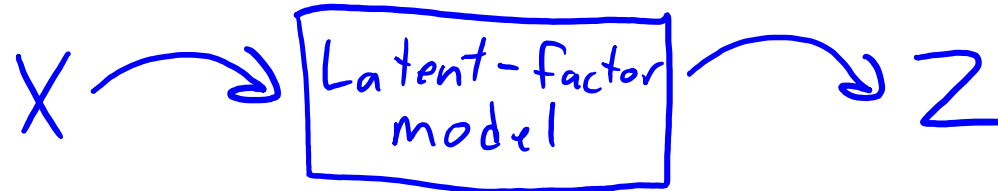


CPSC 340: Machine Learning and Data Mining

More PCA

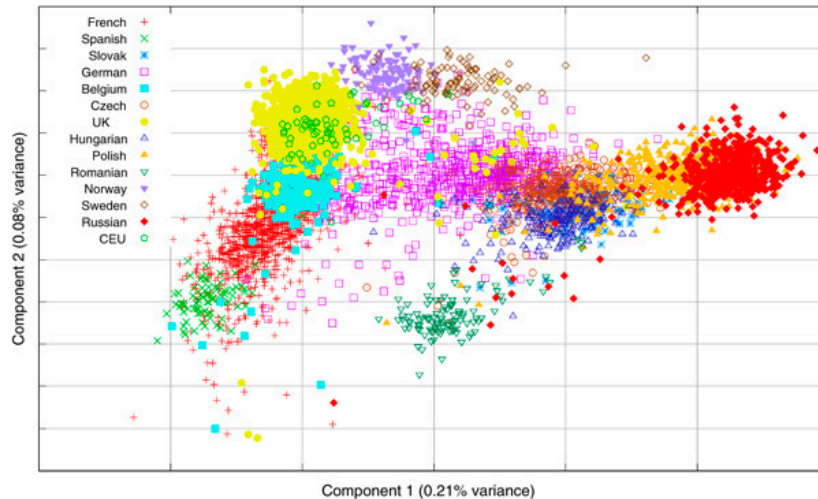
Last Time: Latent-Factor Models

- Latent-factor models take input data 'X' and output a basis 'Z':



بیا، چالب نس

- Usually, 'Z' has fewer features than 'X'.
- Uses: dimensionality reduction, visualization, factor discovery.



Trait	Description
O penness	Being curious, original, intellectual, creative, and open to new ideas.
C onscientiousness	Being organized, systematic, punctual, achievement-oriented, and dependable.
E xtraversion	Being outgoing, talkative, sociable, and enjoying social situations.
A greeableness	Being affable, tolerant, sensitive, trusting, kind, and warm.
N euroticism	Being anxious, irritable, temperamental, and moody.

Last Time: Principal Component Analysis

- Principal component analysis (PCA) is a linear latent-factor model:
 - These models “factorize” matrix X into matrices Z and W :

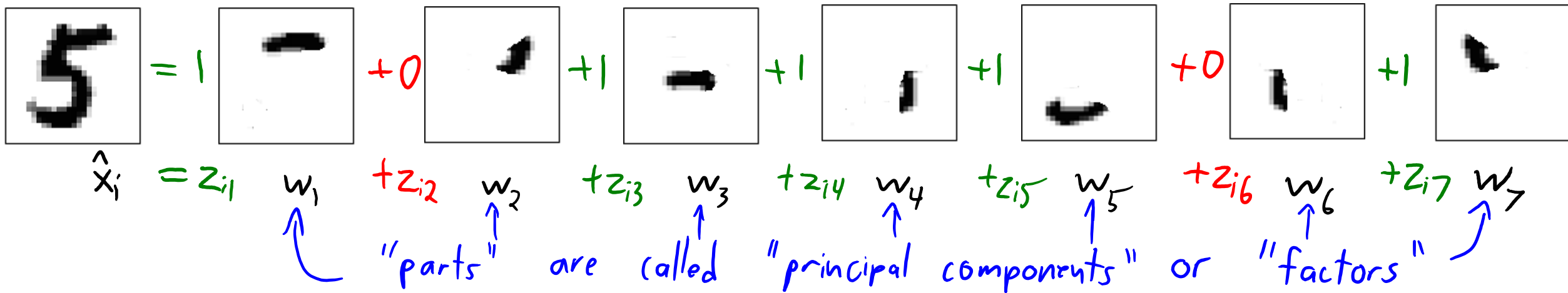
$$X \approx ZW$$

$n \times d$ $n \times k$ $k \times d$

$$x_i \approx W^T z_i$$

$$x_{ij} \approx \langle w_j^i, z_i \rangle$$

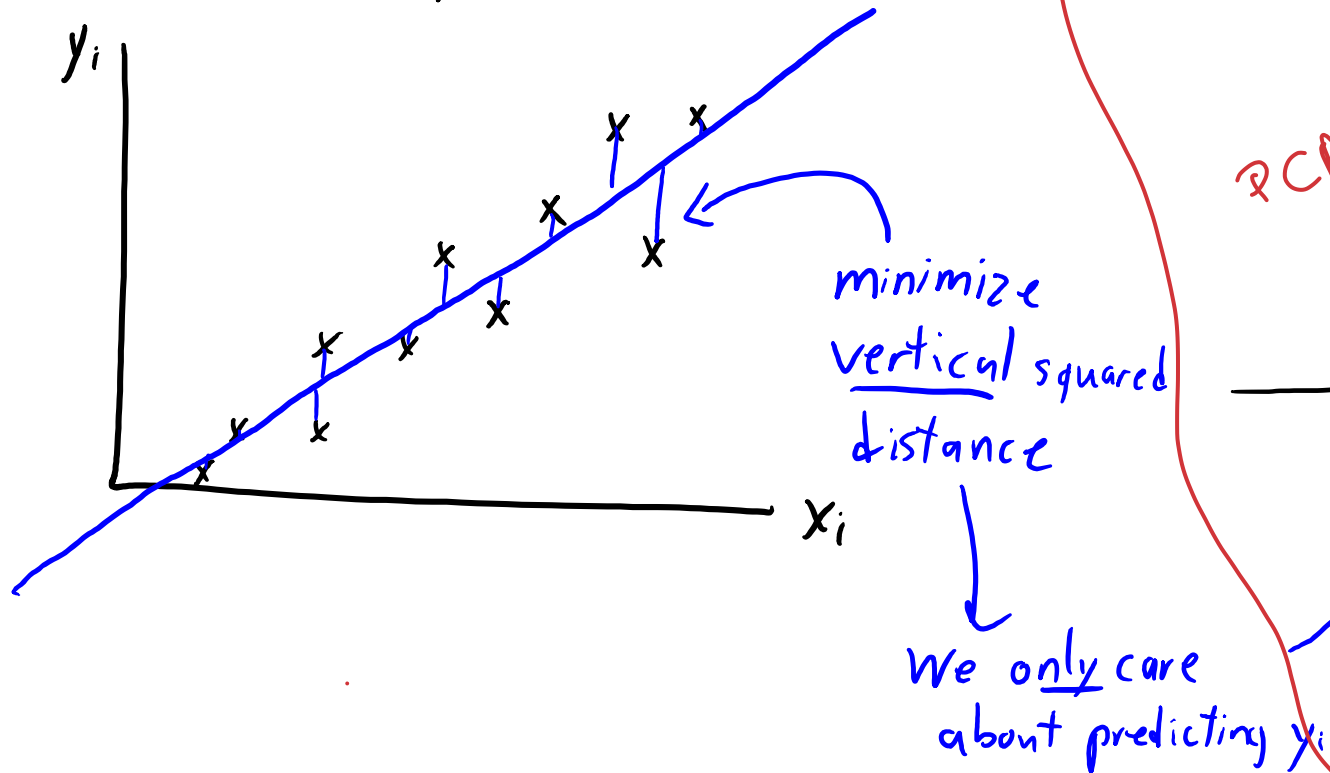
- We can think of rows w_c of W as ‘ k ’ fixed “parts” (used in all examples).
- z_i is the “part weights” for example x_i : “how much of each part w_c to use”.



Next Topic: PCA Geometry in Low Dimensions

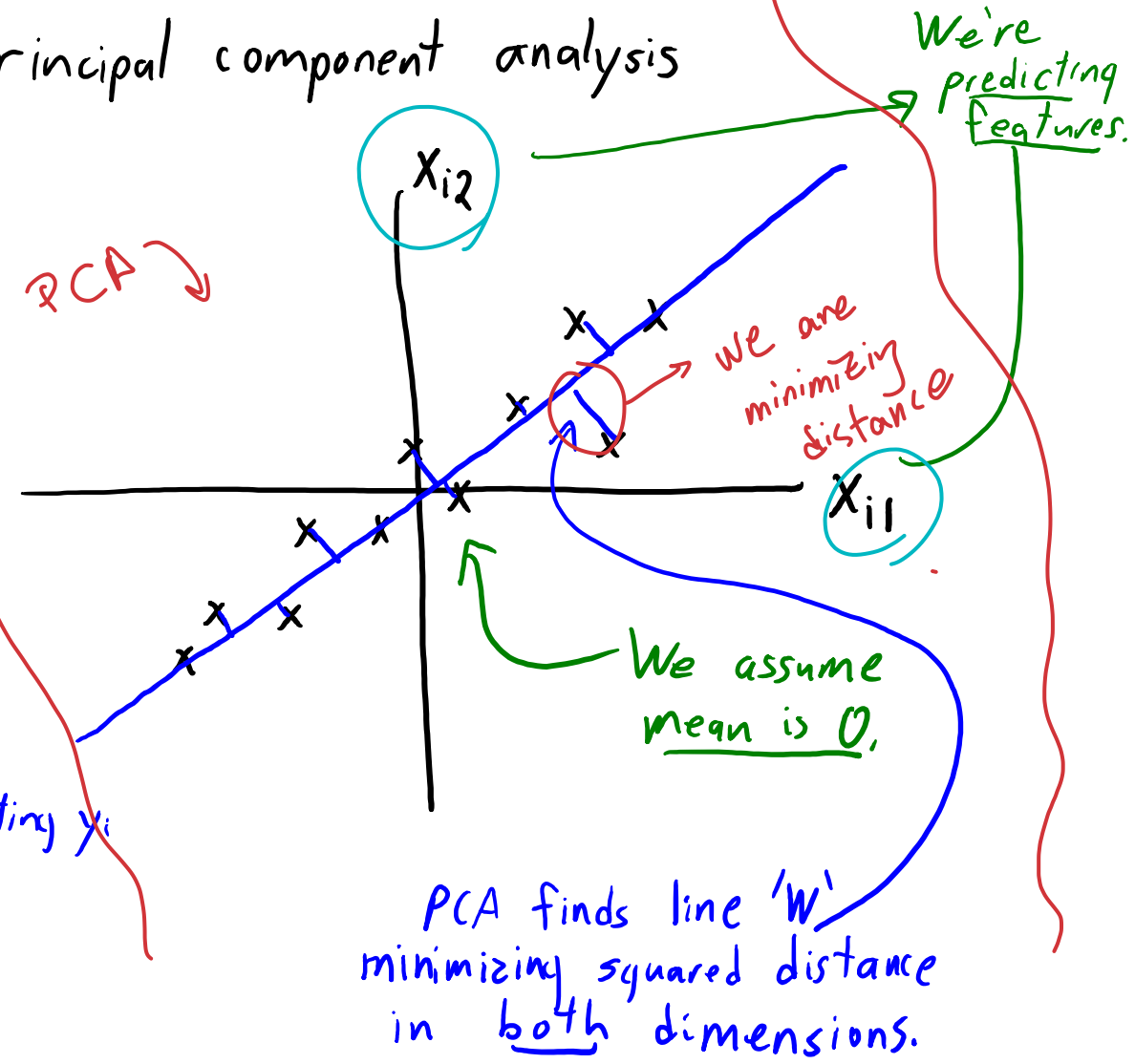
PCA with $d=2$ and $k=1$

Least squares



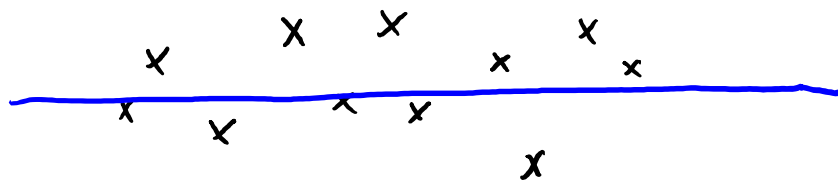
Principal component analysis

PCA →

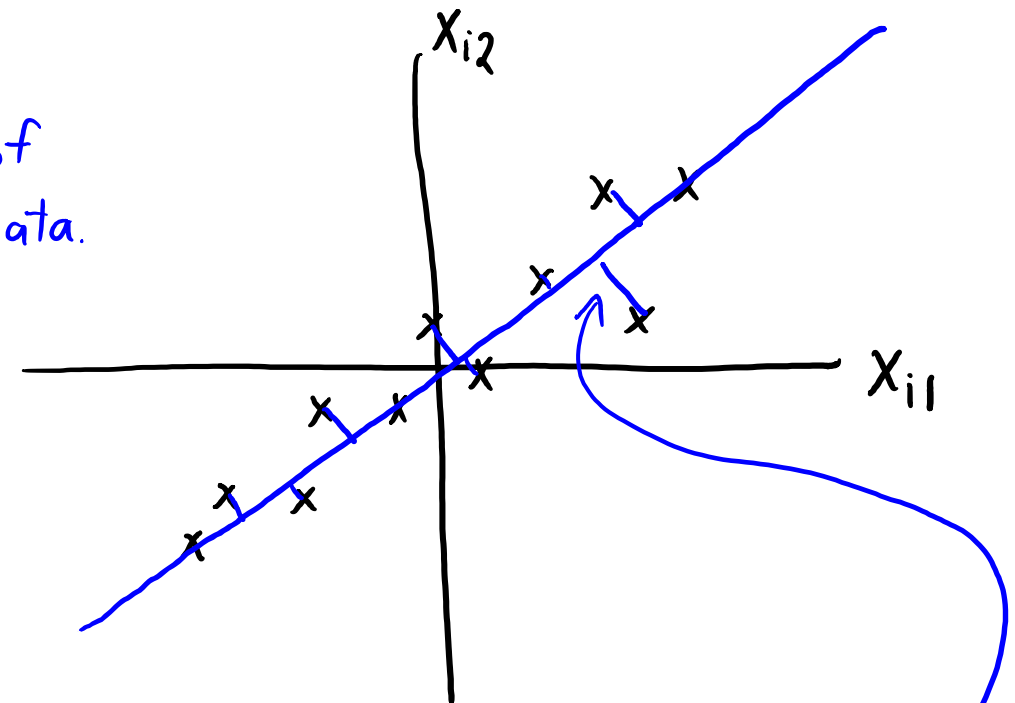


PCA with $d=2$ and $k=1$

Principal component analysis



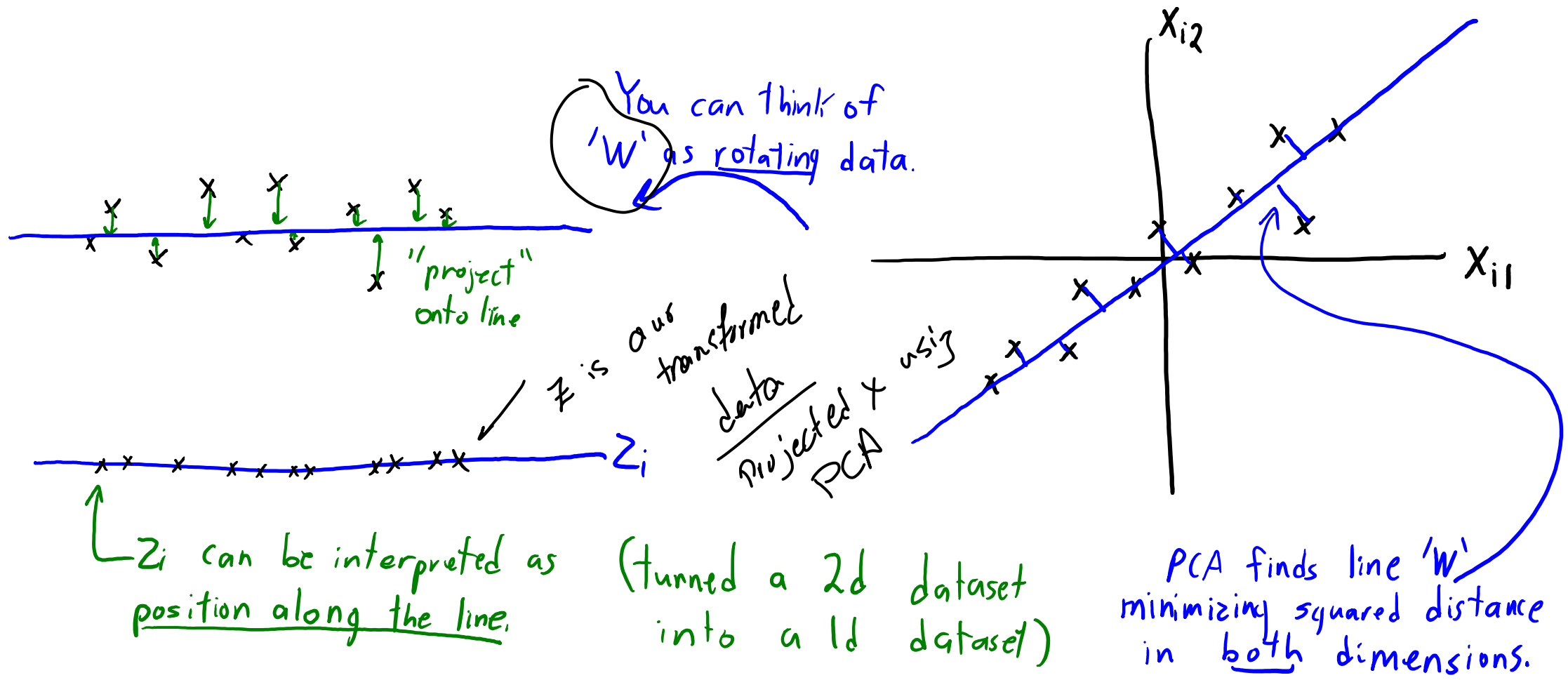
You can think of
'W' as rotating data.



PCA finds line 'W'
minimizing squared distance
in both dimensions.

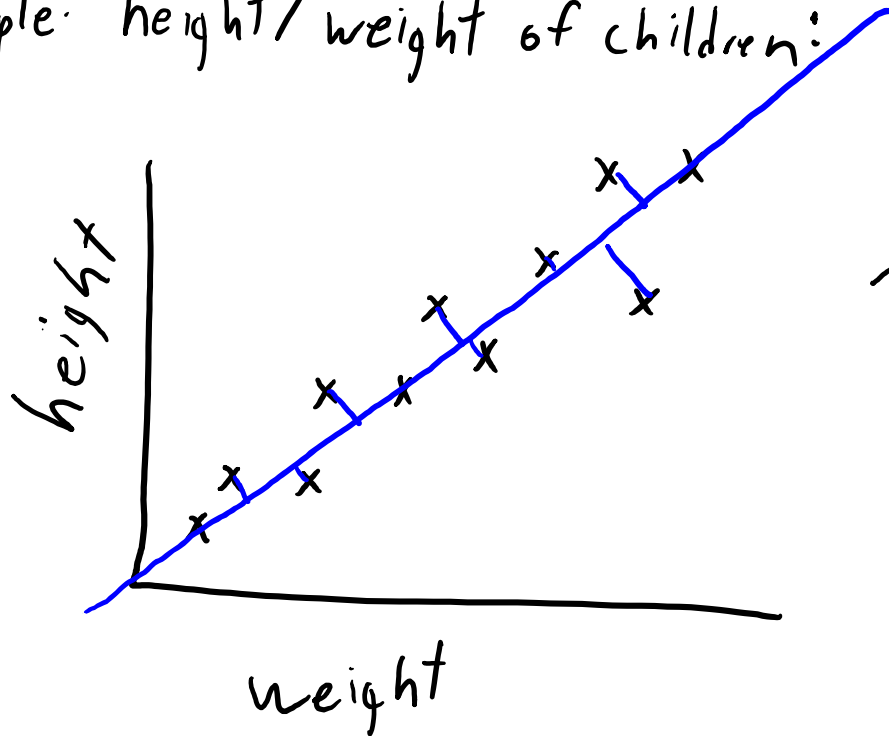
PCA with $d=2$ and $k=1$

Principal component analysis



PCA with $d=2$ and $k=1$

Example: height/weight of children:



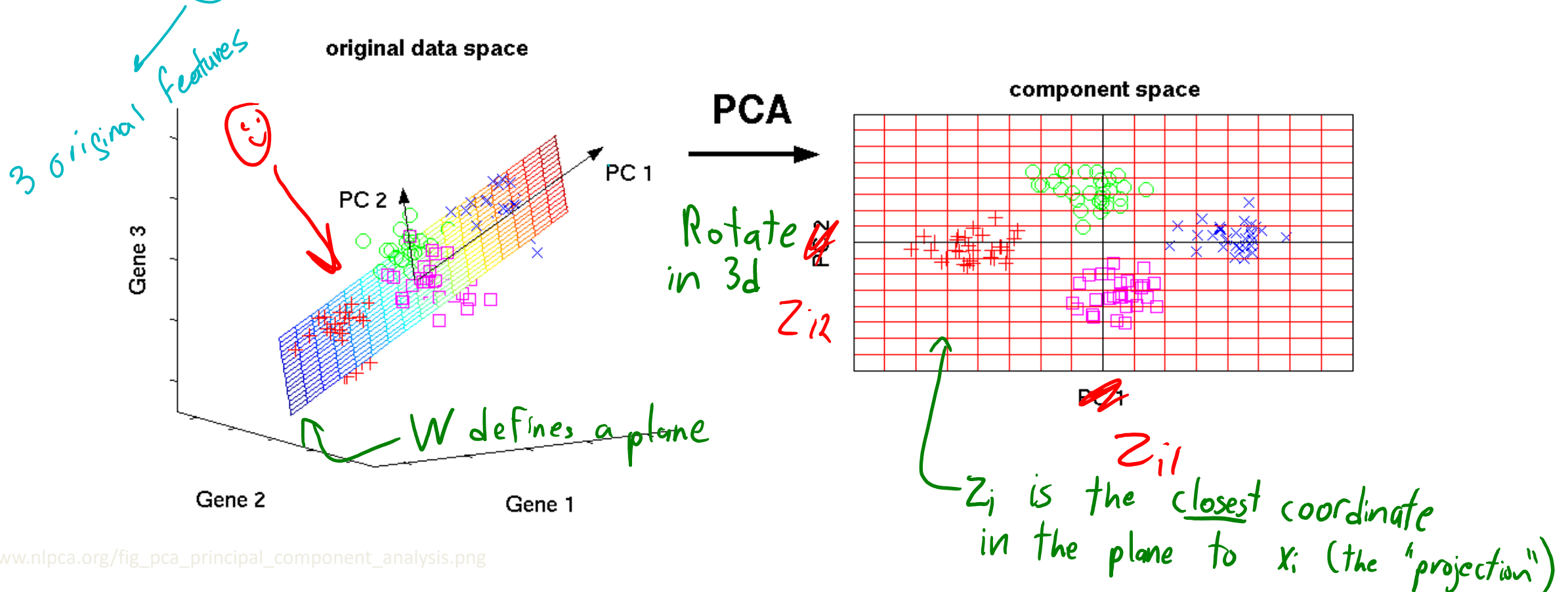
PCA with $k=1$



Latent factor could be viewed as measure of size.

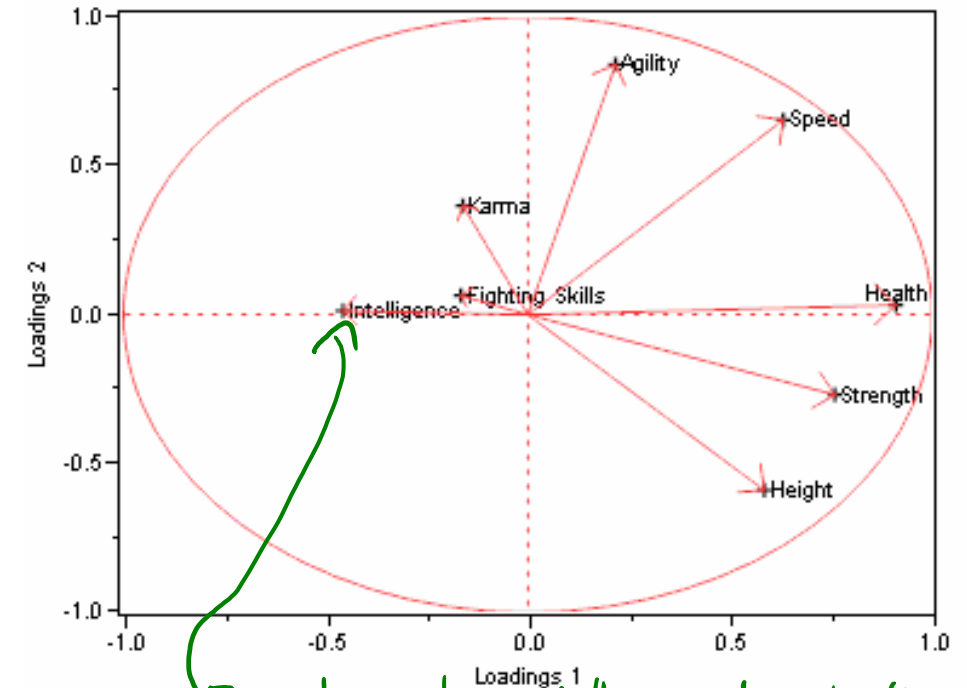
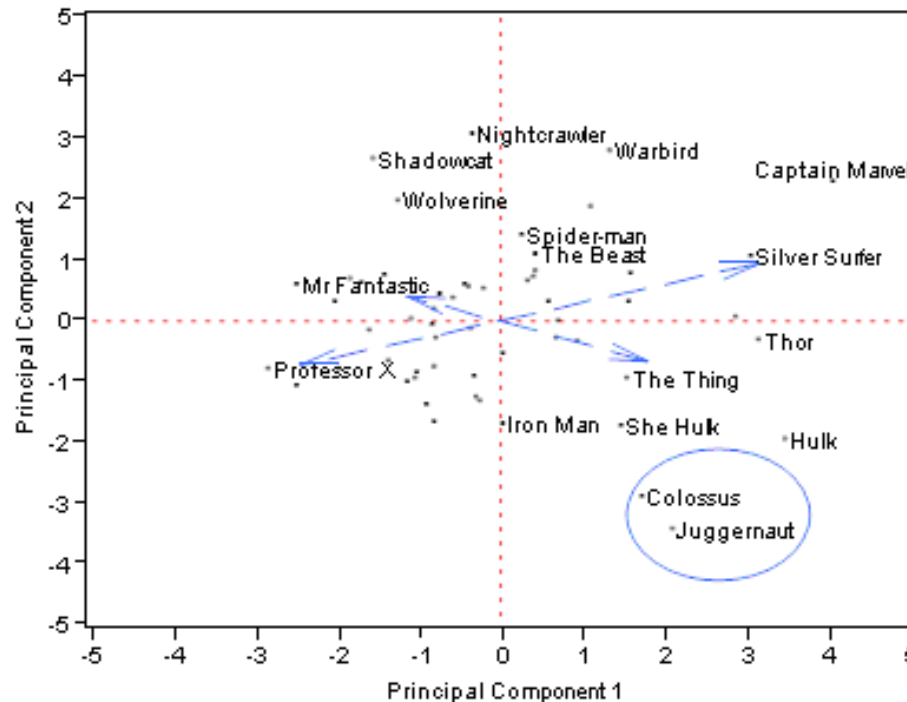
PCA with $d=3$ and $k=2$.

- With $d=3$, PCA ($k=1$) finds line minimizing squared distance to x_i .
- With $d=3$, PCA ($k=2$) finds plane minimizing squared distance to x_i .



PCA Geometry with any 'd' and k=2

- With any 'd', when $k=2$ the W matrix defines a plane:
 - Even if the original data is high-dimensional, we can visualize data “projected” onto this plane.



Next Topic: PCA Loss Function and Prediction

PCA Objective Function

- In PCA we minimize the squared error of the approximation:

$$f(W, z) = \sum_{i=1}^n \| \underbrace{W^T z_i}_{\text{approximation}} - \underbrace{x_i}_{\text{example } i} \|^2$$

Projection to Plane

$W^T z_i \approx x_i$
 $x_i \approx Z W$
 $W^T z_i \approx \langle z_i, w_j \rangle$?

- This is equivalent to the k-means objective:
 - In k-means z_i only has a single '1' value and other entries are zero.

mean for that center
- But in PCA, z_i can be any real number.
 - We approximate x_i as a linear combination of all means/factors.

PCA Objective Function

- In PCA we minimize the **squared error of the approximation**:

$$f(W, Z) = \sum_{i=1}^n ||\underbrace{W^T}_{\text{approximation}} \underbrace{z_i}_{\text{example 'i'}} - \underbrace{x_i}_{\text{example 'i'}}||^2 = \sum_{i=1}^n \sum_{j=1}^d (\underbrace{\langle w^j, z_i \rangle}_{\text{approximation}} - \underbrace{x_{ij}}_{\text{feature 'j' of example 'i'}})^2$$

- We can also view this as **solving 'd' regression problems**:

- Each w^j is trying to predict column ' x^j ' from the basis z_i .

- The output " y_i " we try to predict here is actually the features " x_i ".

- And unlike in regression we are also **learning the features z_i** .

Both w and z
are unknown
and we are
predicting
them.

Principal Component Analysis (PCA)

- The 3 different ways to write the **PCA objective function**:

$$f(W, Z) = \sum_{i=1}^n \sum_{j=1}^d (\langle w^j, z_i \rangle - x_{ij})^2$$

(approximating x_{ij} by $\langle w^j, z_i \rangle$)

$$= \sum_{i=1}^n \|W^T z_i - x_i\|^2$$

(approximating x_i by $W^T z_i$)

$$= \|ZW - X\|_F^2$$

(approximating X by ZW)

Forbinner's
Norm.

Digression: Data Centering (Important)

- In PCA, we assume that the data X is “centered”.
 - Each column of X has a mean of zero.
- It's easy to center the data:

$$\text{Set } \mu_j = \frac{1}{n} \sum_{i=1}^n x_{ij} \quad (\text{mean of column 'j'})$$

Replace each x_{ij} with $(x_{ij} - \mu_j)$

- There are PCA variations that estimate “bias in each coordinate”.
 - In basic model this is equivalent to centering the data.

PCA Computation: Prediction

- At the end of training, the “model” is the μ_j and the W matrix.
 - PCA is parametric.
- PCA prediction phase:
 - Given new data $\tilde{X}$, we can use μ_j and W this to form $\tilde{Z}$:

1. Center: replace each $\tilde{x}_{ij}$ with $(\tilde{x}_{ij} - \mu_j)$

2. Find $\tilde{Z}$ minimizing squared error:

$$\tilde{Z} = \tilde{X} W^T (W W^T)^{-1}$$

(could just store
this $d \times k$ matrix)

means of
training
data

PCA Computation: Prediction

- At the end of training, the “model” is the μ_j and the W matrix.
 - PCA is parametric.
- PCA prediction phase:
 - Given new data $\tilde{X}$, we can use μ_j and W this to form $\tilde{Z}$:
 - The “reconstruction error” is how close approximation is to $\tilde{X}$:

$$\| \underbrace{\tilde{Z}W}_{\hat{X}} - \tilde{X} \|_F^2$$

↑ centered version

- Our “error” from replacing the x_i with the z_i and W .

Choosing 'k' by "Variance Explained"

- Common to choose 'k' based on **variance of the x_{ij}** .

$$\text{Var}(x_{ij}) = \underbrace{E[(x_{ij} - \underbrace{\mu_{ij}}_{\text{assumed to be zero}})^2]}_{\text{definition of variance}} = \underbrace{E[x_{ij}^2]}_{\text{definition of expectation}} = \frac{1}{nd} \sum_{i=1}^n \sum_{j=1}^d x_{ij}^2 = \frac{1}{nd} \|X\|_F^2$$

$\frac{1}{nd} \|X\|_F^2$ is the **Frobenius norm**.
 (Note: $\frac{1}{nd} \|X\|_F^2$ is also the **variance** of the x_{ij} .)

- For a given 'k' we compute (variance of errors)/(variance of x_{ij}):

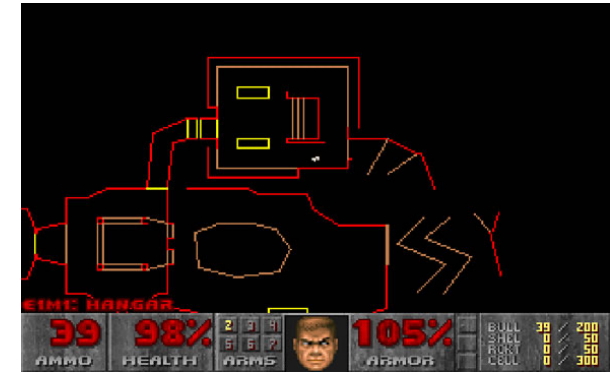
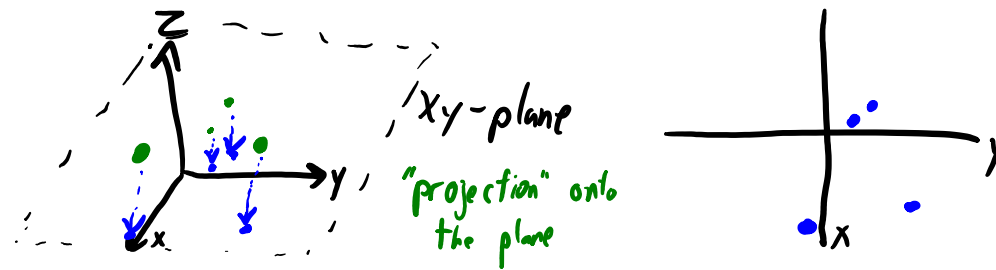
$$\frac{\|ZW - X\|_F^2}{\|X\|_F^2}$$

The numerator $\|ZW - X\|_F^2$ is the **Reconstruction Error**.
 The denominator $\|X\|_F^2$ is the **Centered version** of the variance of x_{ij} .
 The ratio represents the **Remaining Variance**.

- Gives a number between 0 ($k=d$) and 1 ($k=0$), giving "variance remaining".
 - If you want to "explain 90% of variance", choose smallest 'k' where ratio is < 0.10 .

“Variance Explained” in the Doom Map

- Recall the Doom latent-factor model (where map ignores height):



- Interpretation of “variance remaining” formula:

$$\frac{\|ZW - X\|_F^2}{\|X\|_F^2} \leftarrow \begin{array}{l} \text{Variance in } z\text{-dimension (variance in } x\text{- and } y\text{-dimensions fully} \\ \text{captured by overhead map)} \end{array}$$

$$\leftarrow \text{Variance of character in 3-dimensions}$$

- If we had a 3D map the “variance remaining” would be 0.

$X = ZW \rightarrow$ How to make the matrix Projection.
 $x \cdot w$ (dot product between x (vector) here and w)
 $\max \|x \cdot w\|$ for all unit $\|w\|$

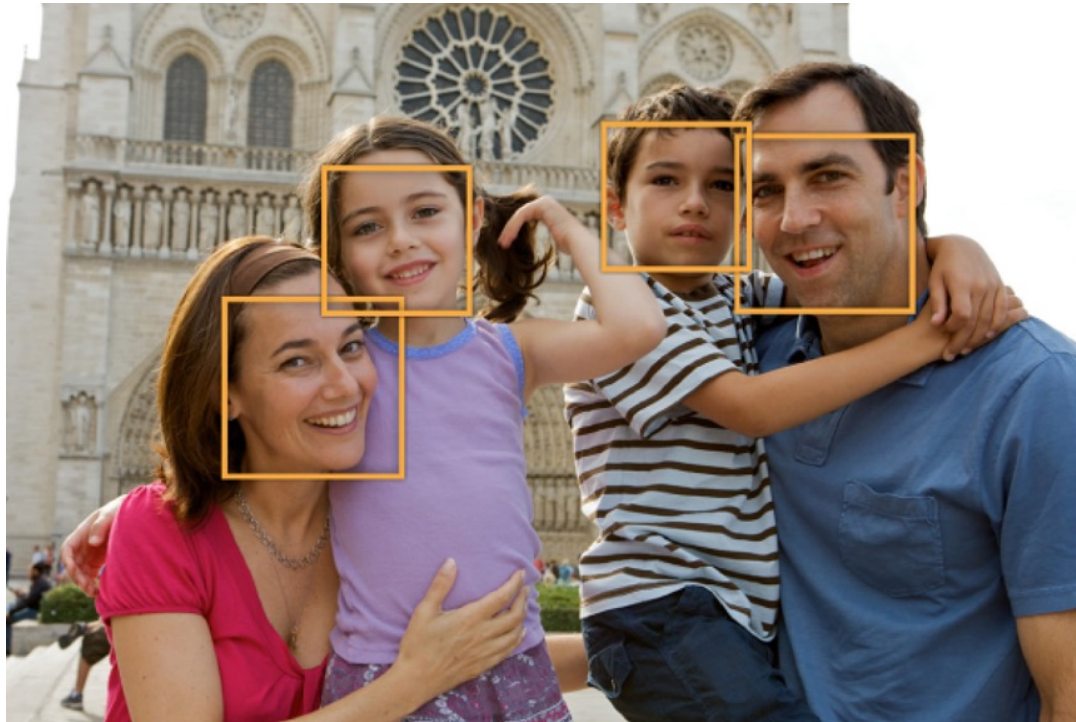
$\rightarrow$ This will be the
eigen vectors of $x^T x$

Next Topic: Eigenfaces

$x \cdot w$ is high if x and w are aligned.
or a GP of it. $\rightarrow$ Eigen vectors of $x^T x$

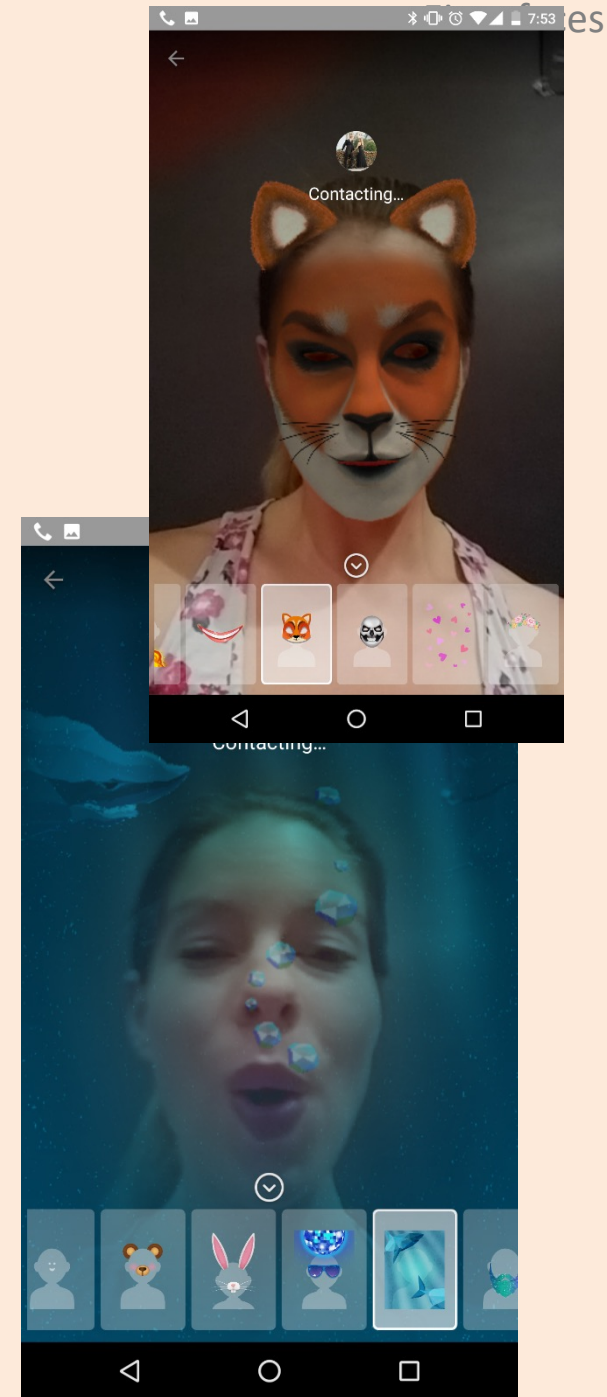
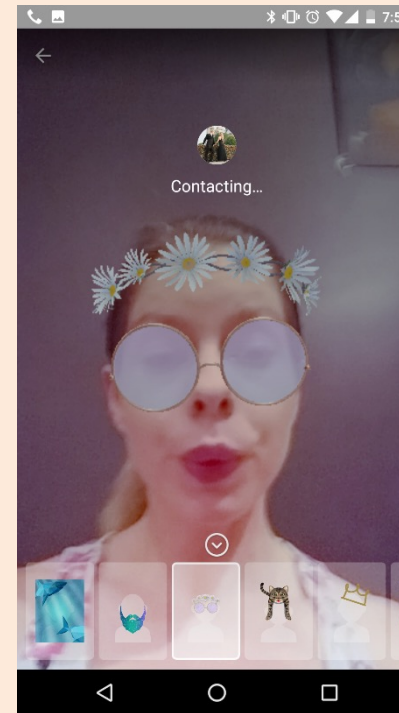
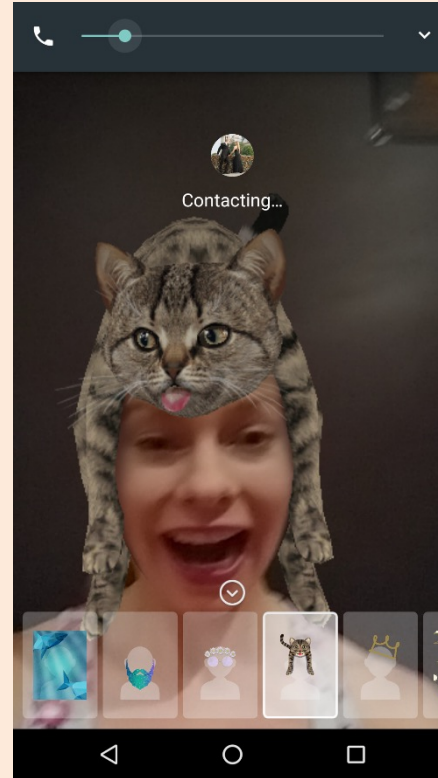
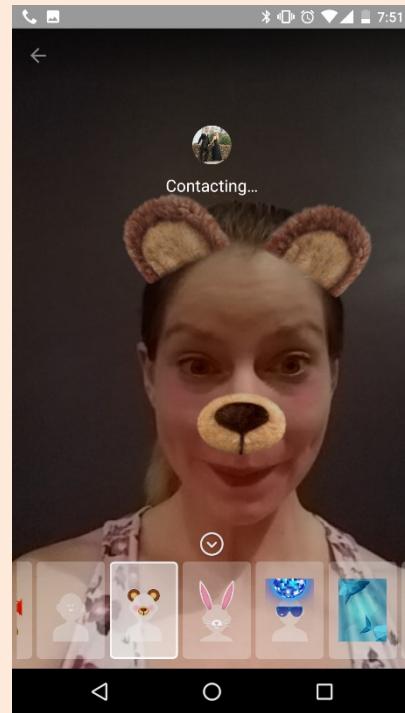
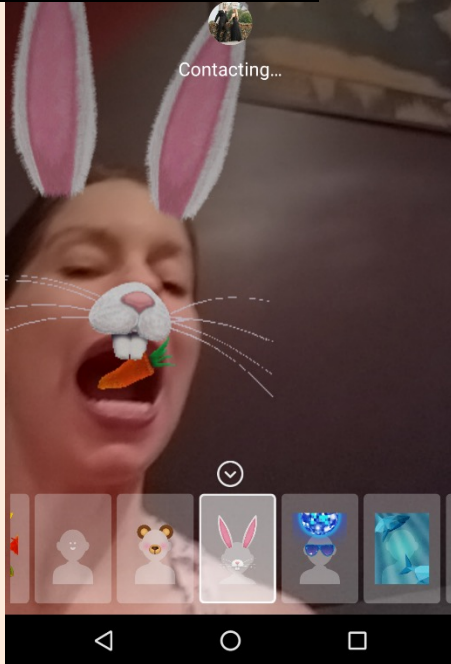
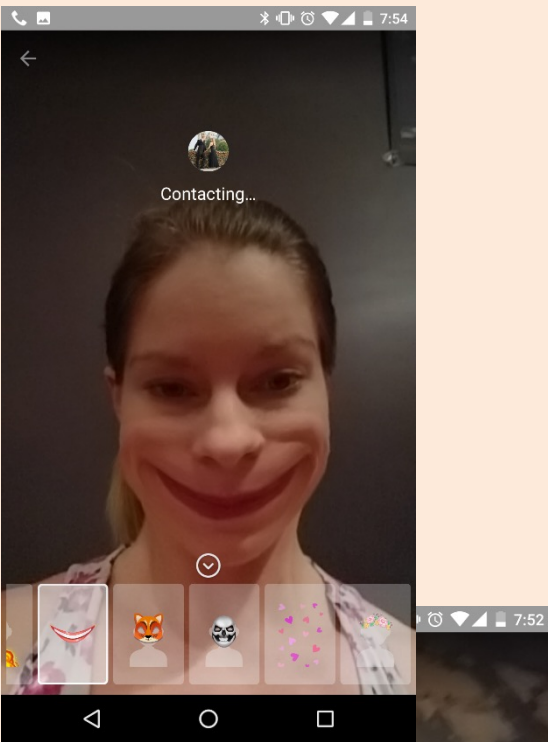
Application: Face Detection

- Consider problem of face detection:



- Classic methods use “eigenfaces” as basis:
 - PCA applied to images of faces.

Application: Face Detection



Eigenfaces

- Collect a bunch of images of faces under different conditions:



Each row of X will be pixels in one image:

$X =$

Flatten
first

the image so that
column will be the first
row of the line on the
next columns....

If have ' n ' images that are ' m ' by ' m ' then X is ' n ' by m^2

Eigenfaces

Compute mean μ_j of each column,



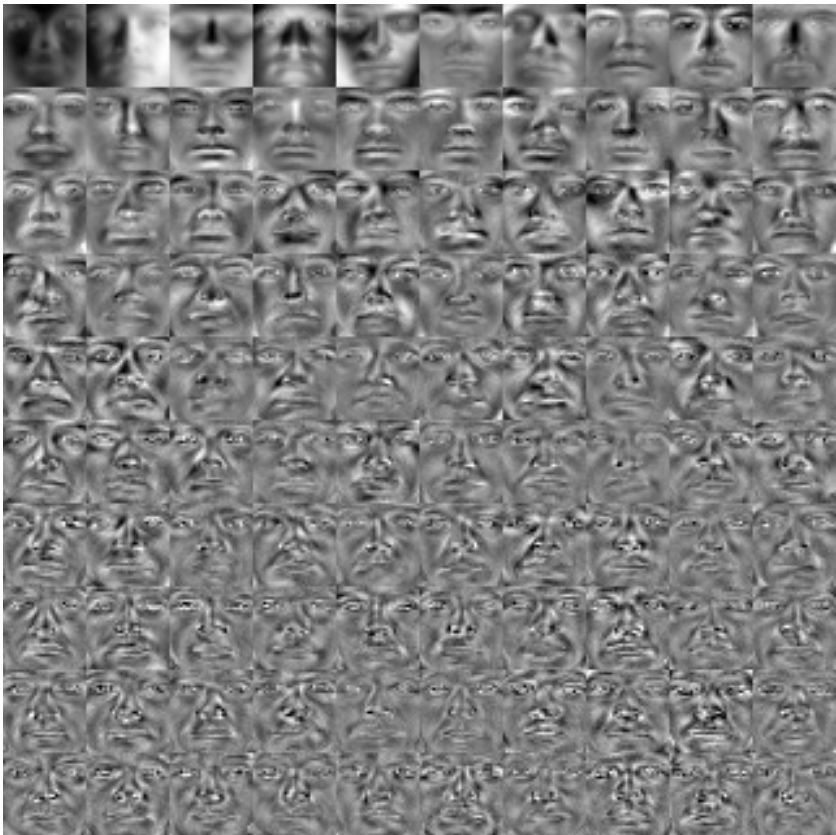
Each row of X will be pixels in one image:

$$X = \begin{bmatrix} \text{---} x_1 - \mu \text{---} \\ \text{---} x_2 - \mu \text{---} \\ \vdots \\ \text{---} x_n - \mu \text{---} \end{bmatrix}$$

Replace each x_{ij} by $x_{ij} - \mu_j$

Eigenfaces

Compute top 'k' PCs on centered data: Each row of X will be pixels in one image:



$$X = \begin{bmatrix} \text{---} x_1 - \mu \text{---} \\ \text{---} x_2 - \mu \text{---} \\ \vdots \\ \text{---} x_n - \mu \text{---} \end{bmatrix}$$

Eigenfaces

Compute top 'k' PCs on centered data:

PC1

PC2

Note that these are "signed" images.

Component 00

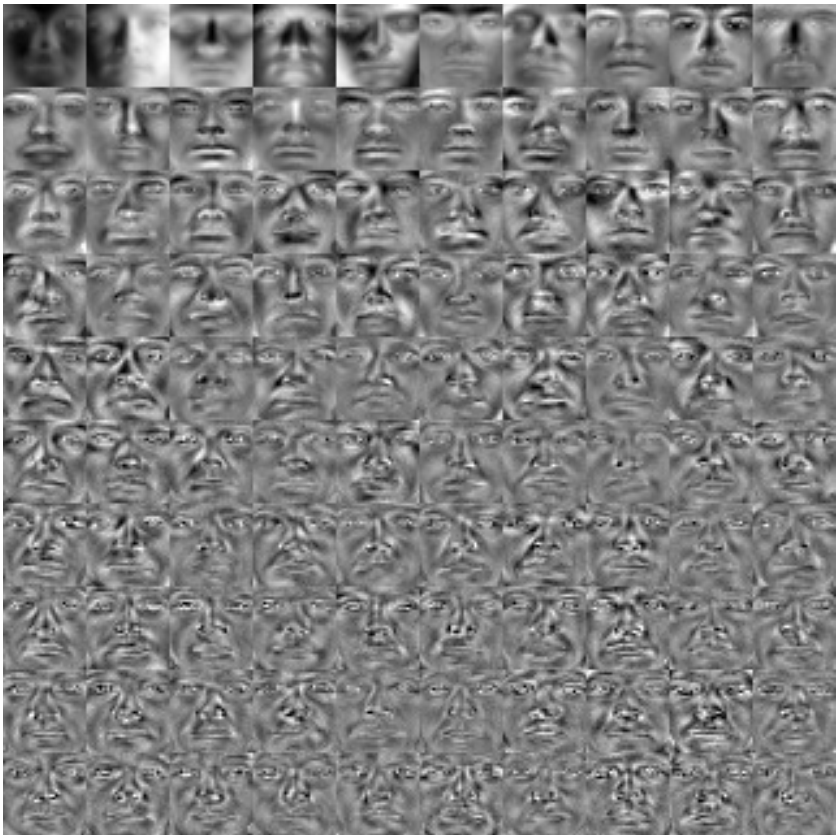
"gray" represents values close to 0.

"dark" represents negative values

"bright" represents positive values

Eigenfaces

Compute top 'k' PCs on centered data:



~~Not a PCA~~

∞%

"Eigenface" representation:

$$\hat{x}_i = \mu + z_{i1} \text{PC1} + z_{i2} \text{PC2} + z_{i3} \text{PC3} + \dots$$

(first row of W)

Eigenfaces

106 of the original faces:



"Eigenface" representation:

$$\hat{x}_i = \mu + z_{i1} \text{PC1} + z_{i2} \text{PC2} + z_{i3} \text{PC3} + \dots$$

$\hat{x}_i$ μ PC1 (first row of W) PC2 PC3

Eigenfaces

Reconstruction with $k=0$



Variance explained: 0%

"Eigenface" representation:

$$\hat{x}_i = \mu + z_{i1} \text{PC1} + z_{i2} \text{PC2} + z_{i3} \text{PC3} + \dots$$

(first row of W)

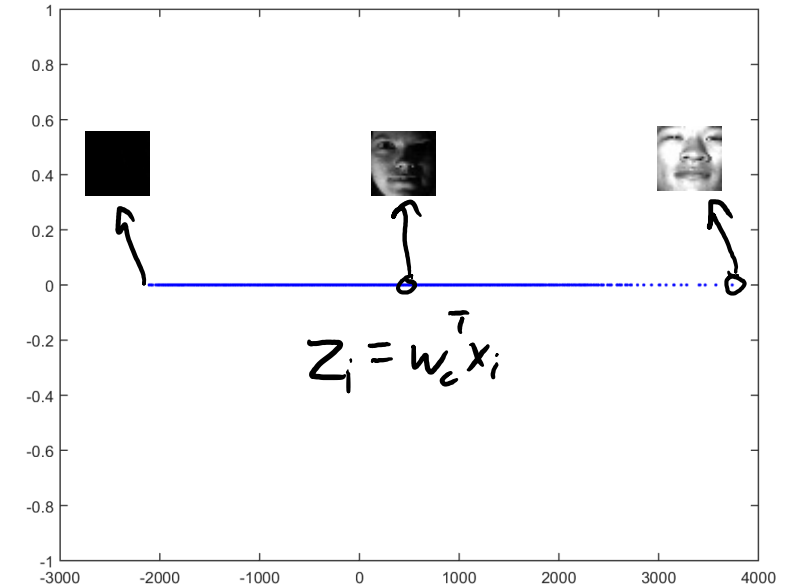
Eigenfaces

Reconstruction with $k=1$



Variance explained: 36%

PCA Visualization:



"Eigenface" representation:

$$\hat{x}_i = \mu + z_{i1} \text{PC1} + z_{i2} \text{PC2} + z_{i3} \text{PC3} + \dots$$

(first row of w)

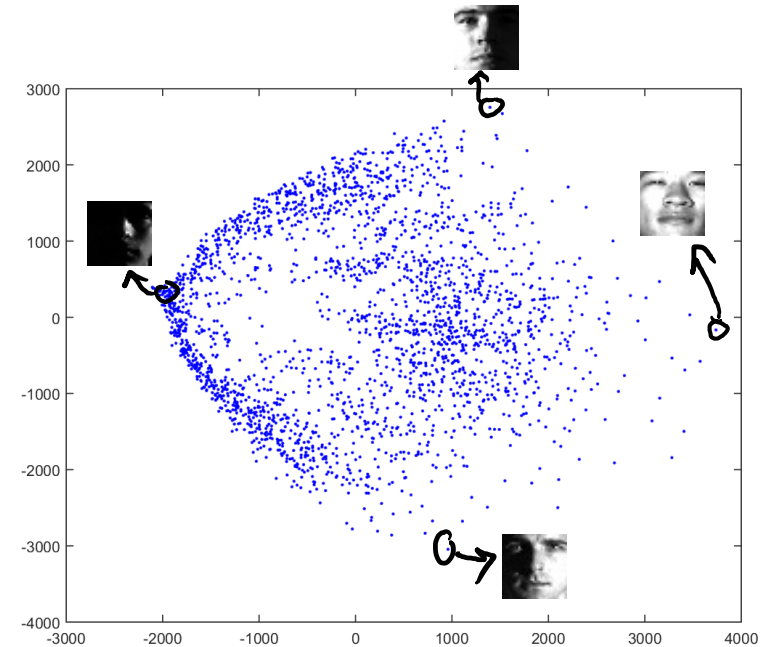
Eigenfaces

Reconstruction with $k=2$



Variance explained: 71%

PCA Visualization:



"Eigenface" representation:

$$\hat{x}_i = \mu + z_{i1} \text{PC1} + z_{i2} \text{PC2} + z_{i3} \text{PC3} + \dots$$

(first row of W)

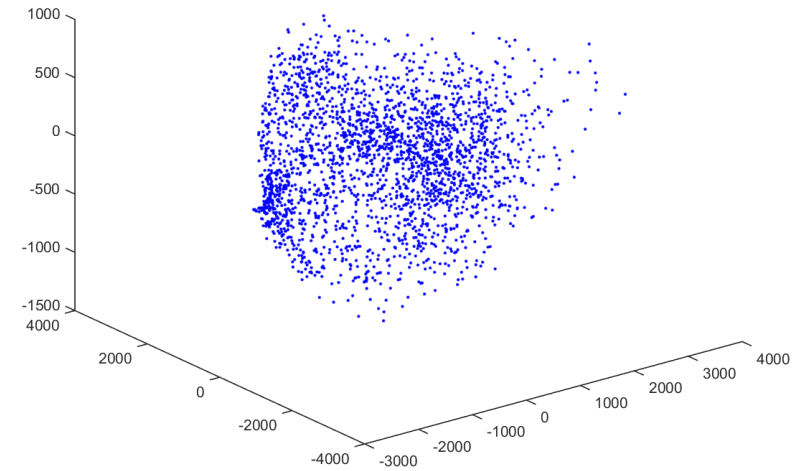
Eigenfaces

Reconstruction with $k=3$



Variance explained: 76%

PCA Visualization:



"Eigenface" representation:

$$\hat{x}_i = \mu + z_{i1} \text{PC1} + z_{i2} \text{PC2} + z_{i3} \text{PC3} + \dots$$

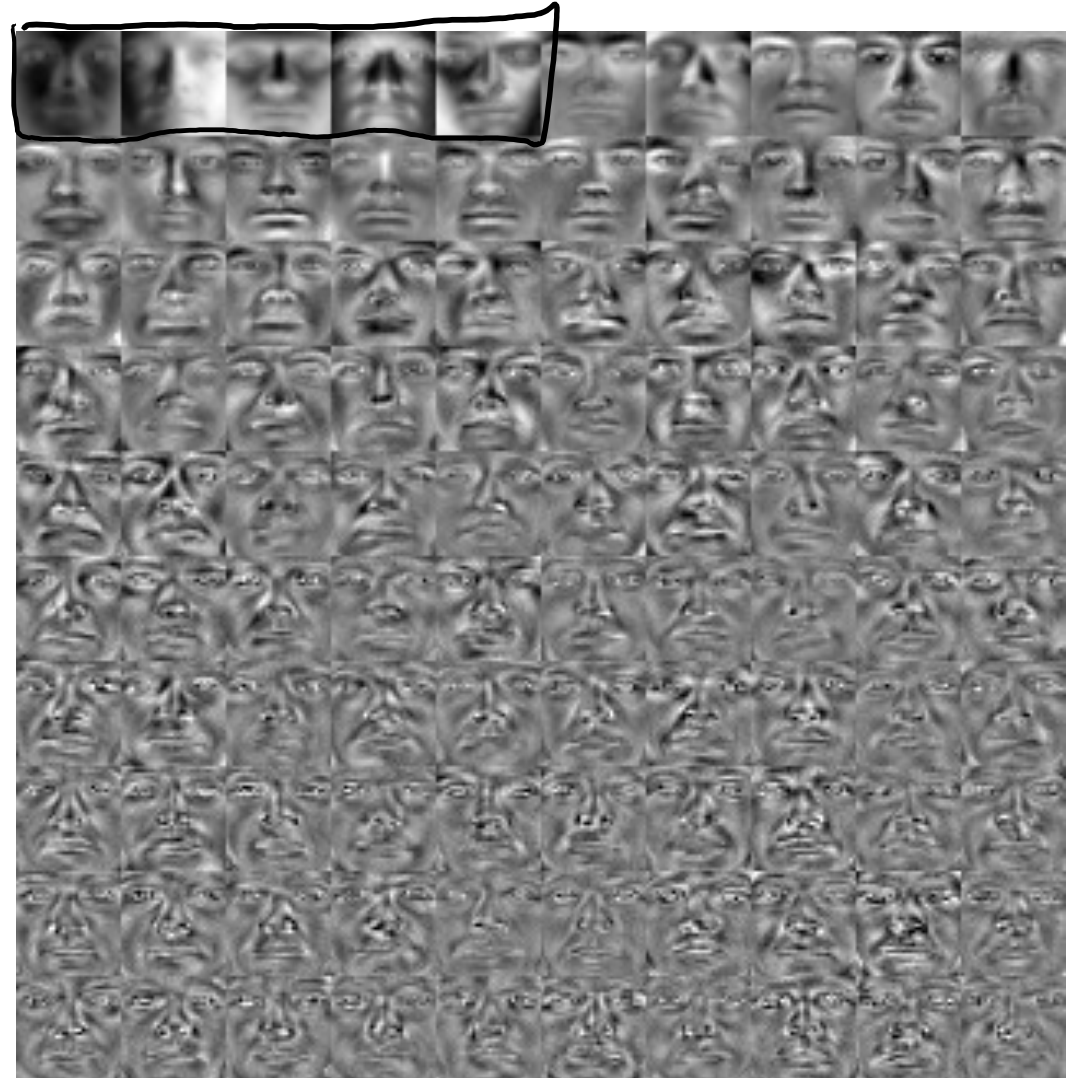
(first row of W)

Eigenfaces

Reconstruction with $k=5$



Variance explained: 86%

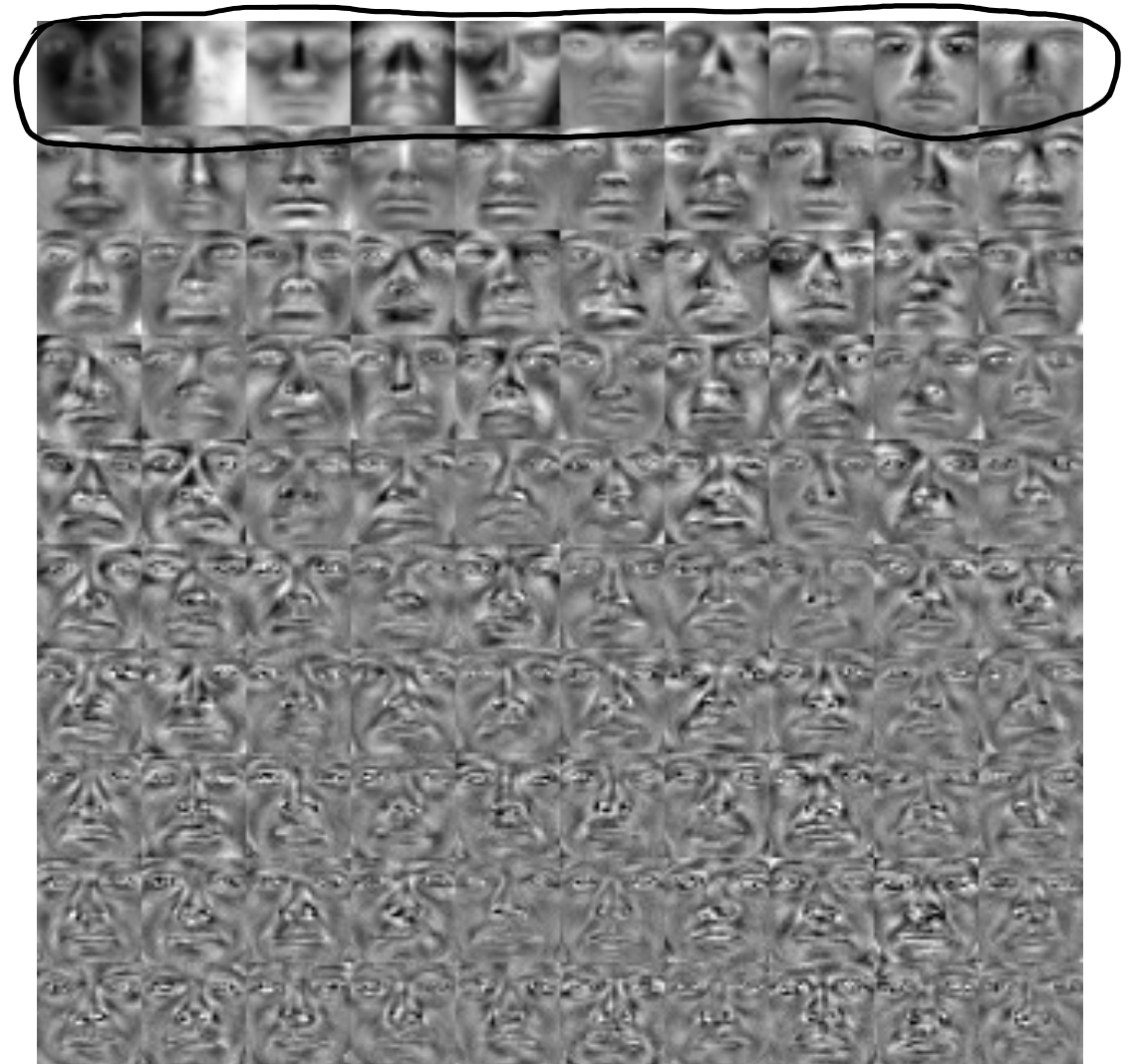


Eigenfaces

Reconstruction with $k=10$



Variance explained: 85%



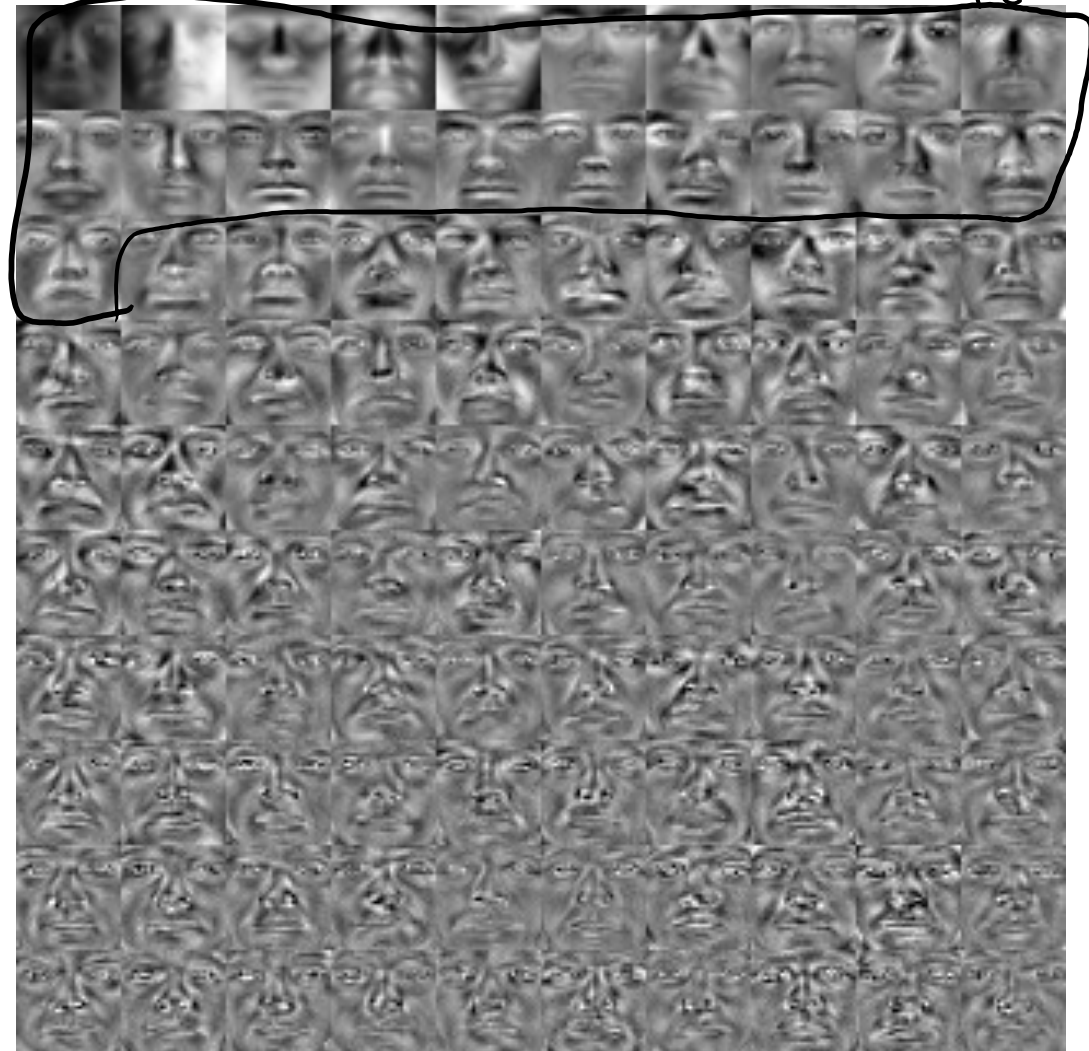
Reconstruction with $k=21$



Variance explained: 90%

Eigenfaces

as if you are only looking at those people

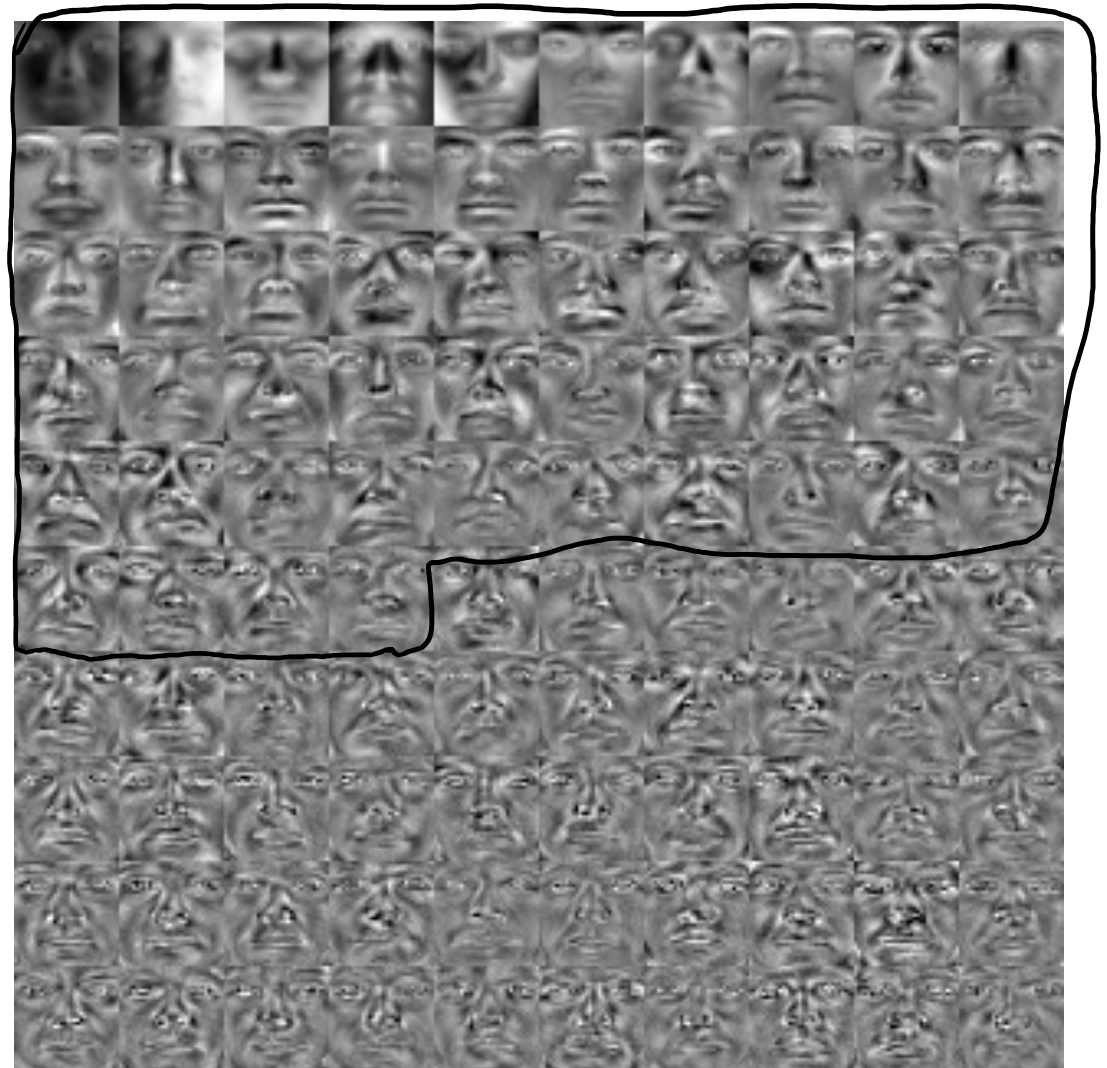


Eigenfaces

Reconstruction with $k=54$



Variance explained: 95%

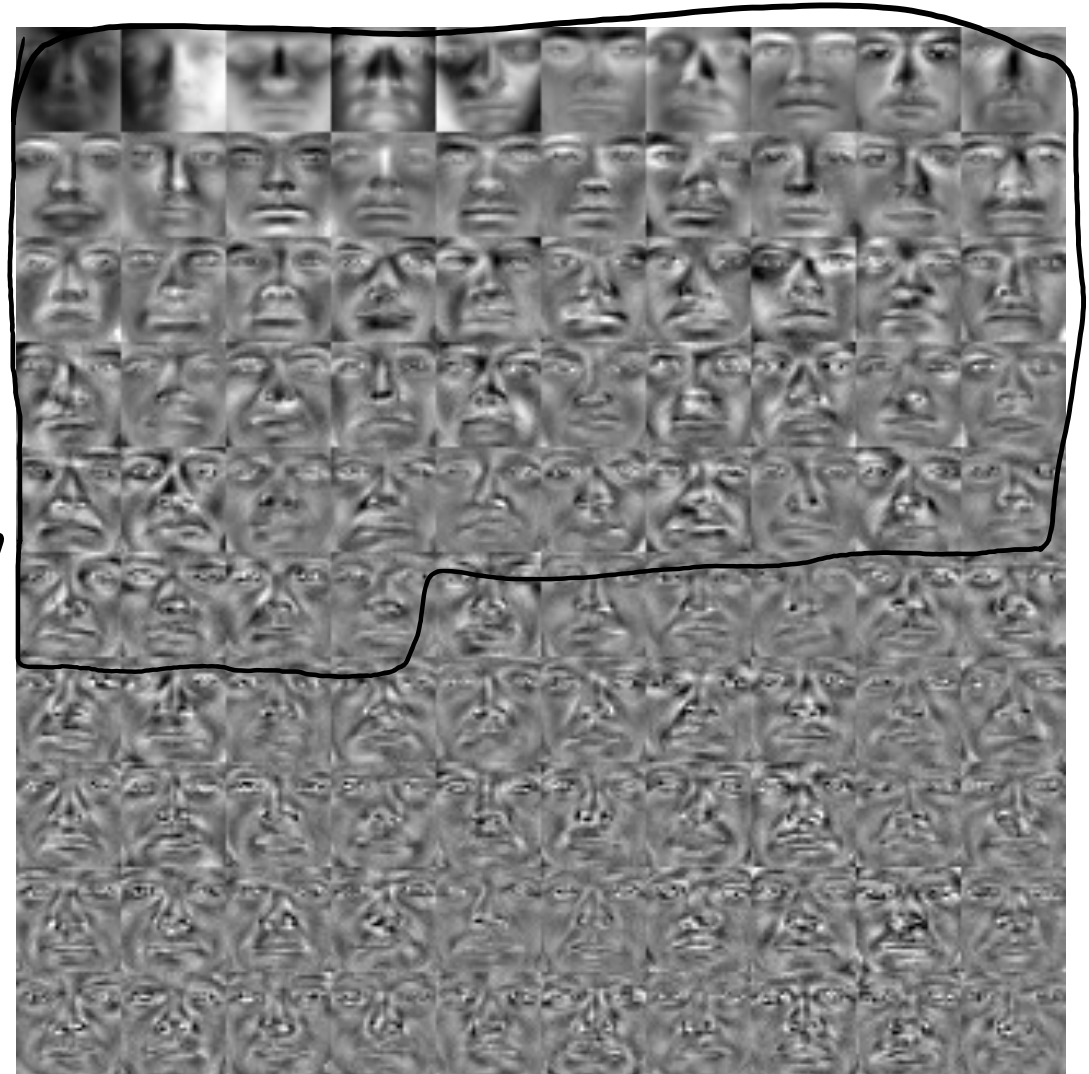


Eigenfaces

Original Images again:



Plus these
"eigenfaces"
and
the
mean.



We can replace 1024 x_i values by 54 z_i values

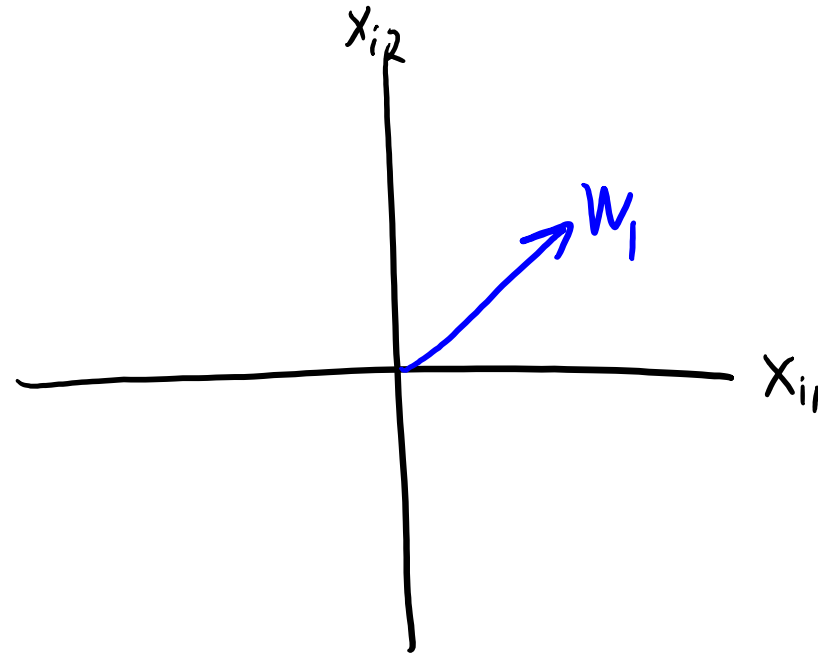
Next Topic: Non-Uniqueness of PCA

Non-Uniqueness of PCA

- Unlike k-means, we **can efficiently find global optima** of $f(W,Z)$.
 - Algorithms coming later.
- Unfortunately, there **never exists a unique global optimum**.
 - There are actually several different sources of non-uniqueness.
- To understand these, we'll need idea of “**span**” from linear algebra.
 - This also helps explain the geometry of PCA.
 - We'll also see that **some global optima may be better than others**.

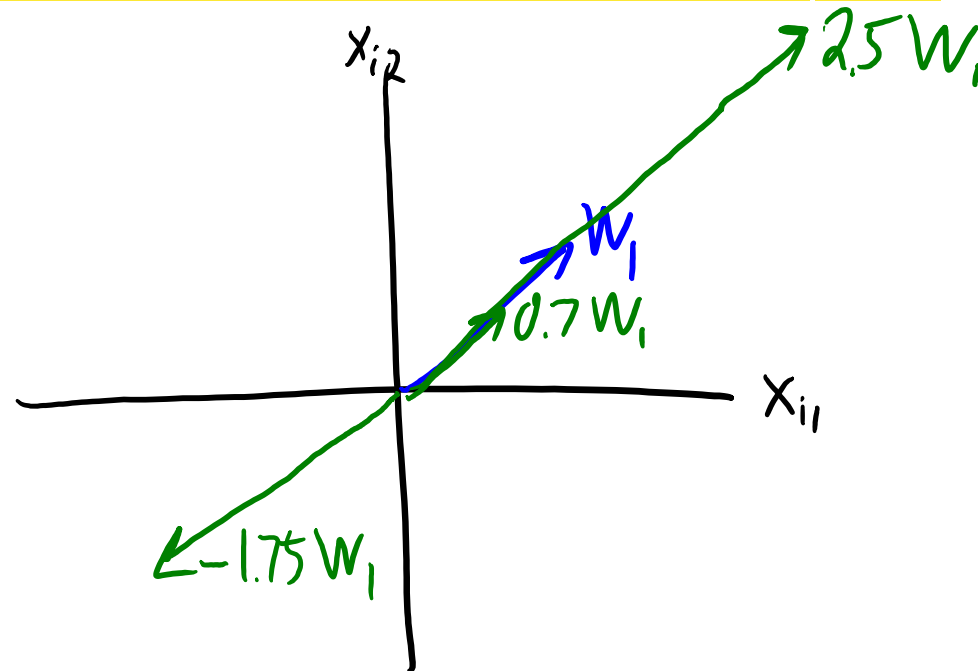
Span of 1 Vector

- Consider a single vector w_1 ($k=1$).



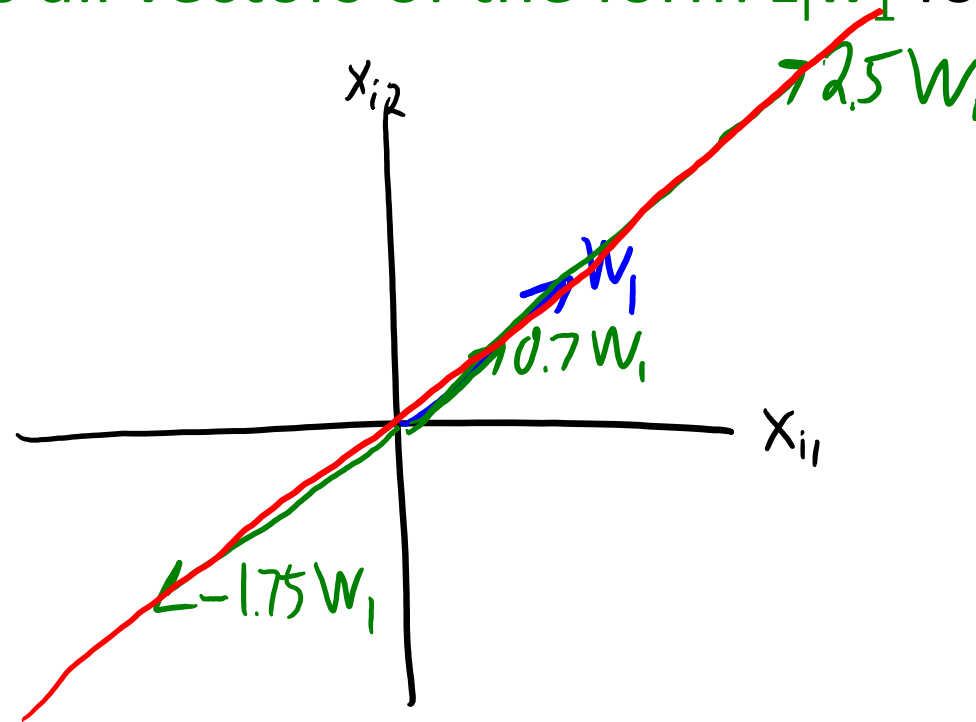
Span of 1 Vector

- Consider a single vector w_1 ($k=1$).
- The $\text{span}(w_1)$ is all vectors of the form $z_i w_1$ for a scalar z_i .



Span of 1 Vector

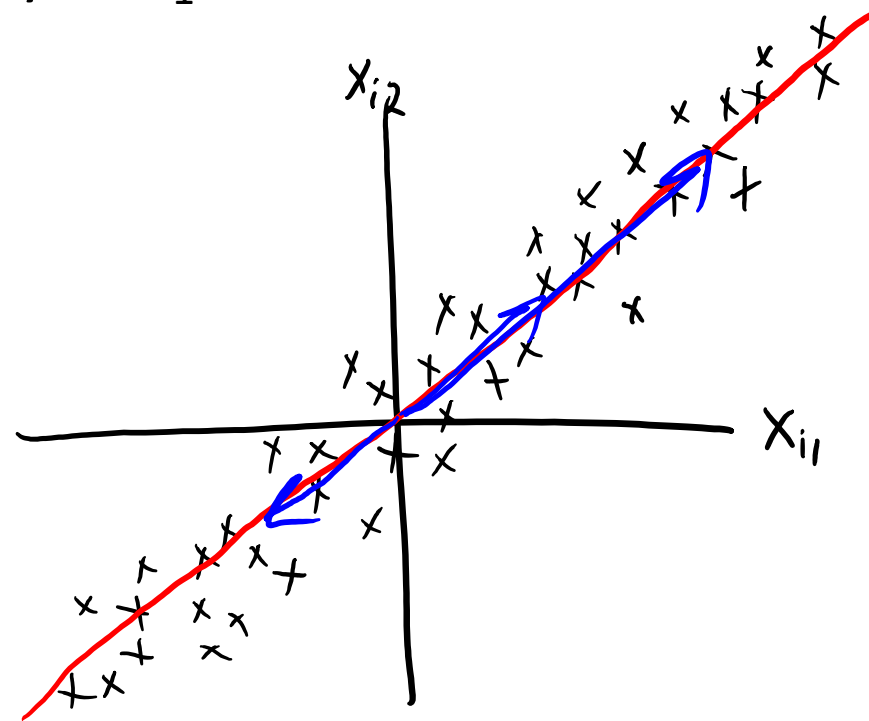
- Consider a single vector w_1 ($k=1$).
- The $\text{span}(w_1)$ is all vectors of the form $z_i w_1$ for a scalar z_i .



- If $w_1 \neq 0$, this forms a line.

Span of 1 Vector

- But note that the “span” of many different vectors gives same line.
 - Mathematically: αw_1 defines the same line as w_1 for any scalar $\alpha \neq 0$.

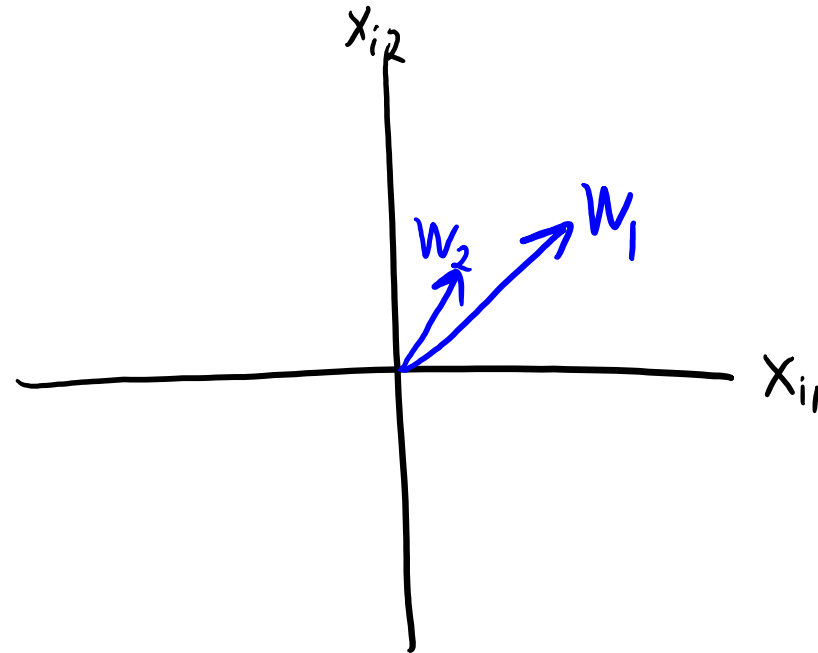


– PCA solution can only be defined up to scalar multiplication.

- If (W, Z) is a solution, then $(\alpha W, (1/\alpha)Z)$ is also a solution. $\|(\alpha W)(\frac{1}{\alpha}Z) - X\|_F^2 = \|WZ - X\|_F^2$

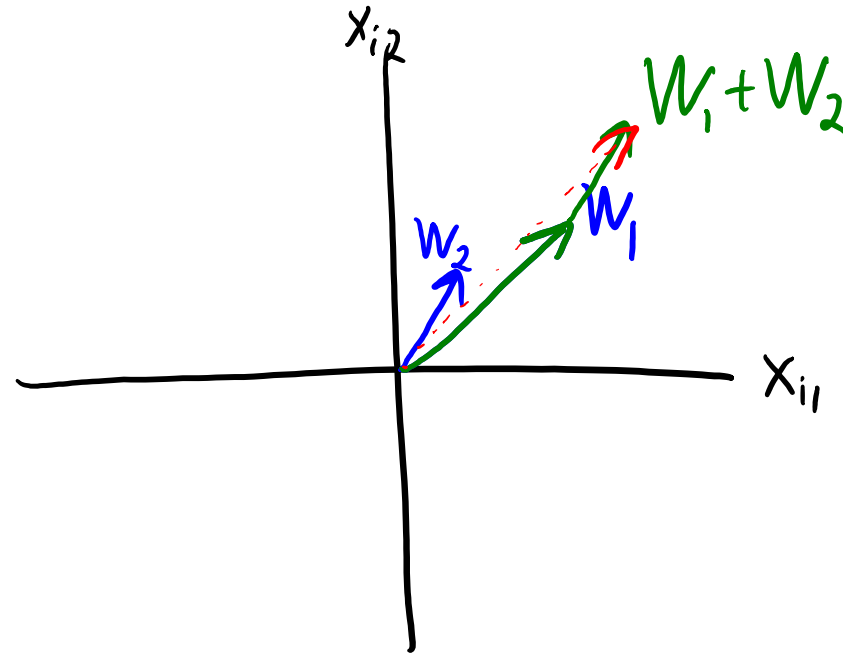
Span of 2 Vectors

- Consider two vector w_1 and w_2 ($k=2$).



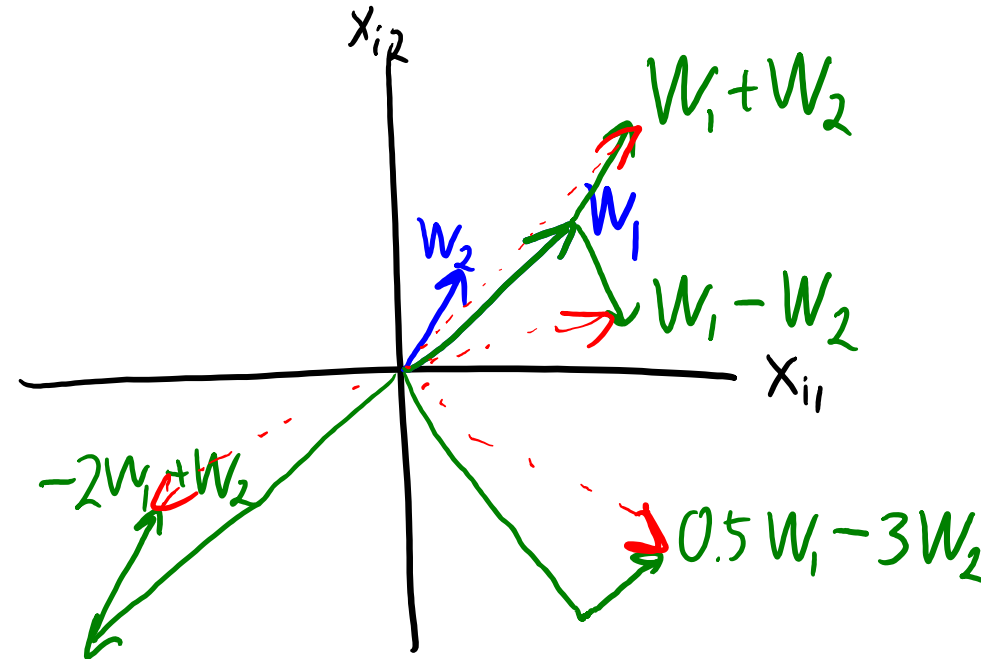
Span of 2 Vectors

- Consider two vector w_1 and w_2 ($k=2$).
 - The $\text{span}(w_1, w_2)$ is all vectors of form $z_{i1}w_1 + z_{i2}w_2$ for a scalars z_{i1} and z_{i2} .



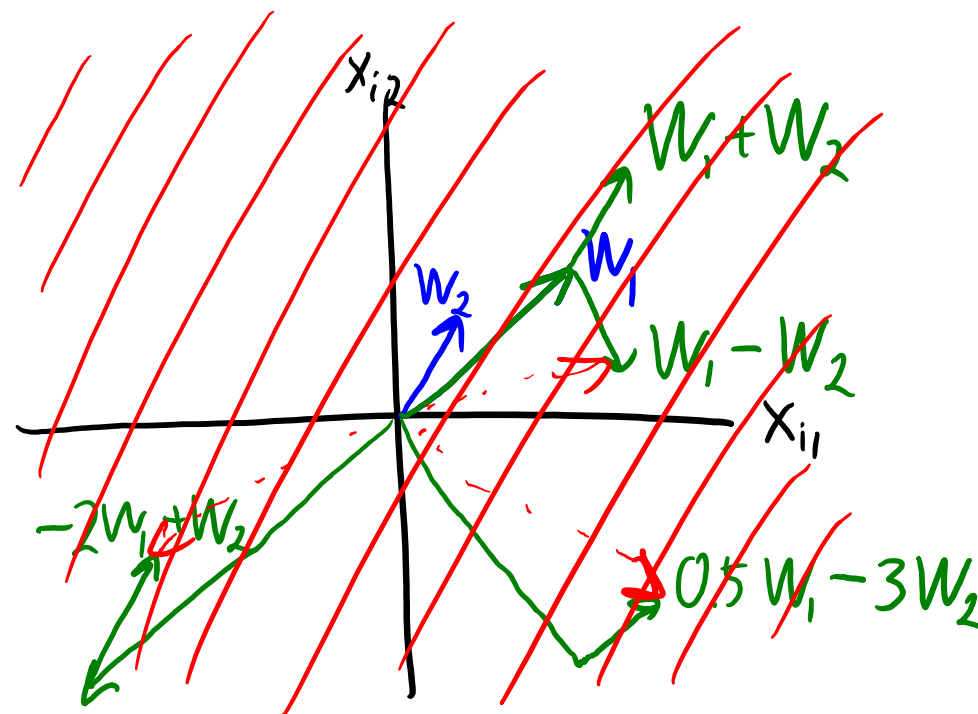
Span of 2 Vectors

- Consider two vector w_1 and w_2 ($k=2$).
 - The $\text{span}(w_1, w_2)$ is all vectors of form $z_{i1}w_1 + z_{i2}w_2$ for a scalars z_{i1} and z_{i2} .



Span of 2 Vectors

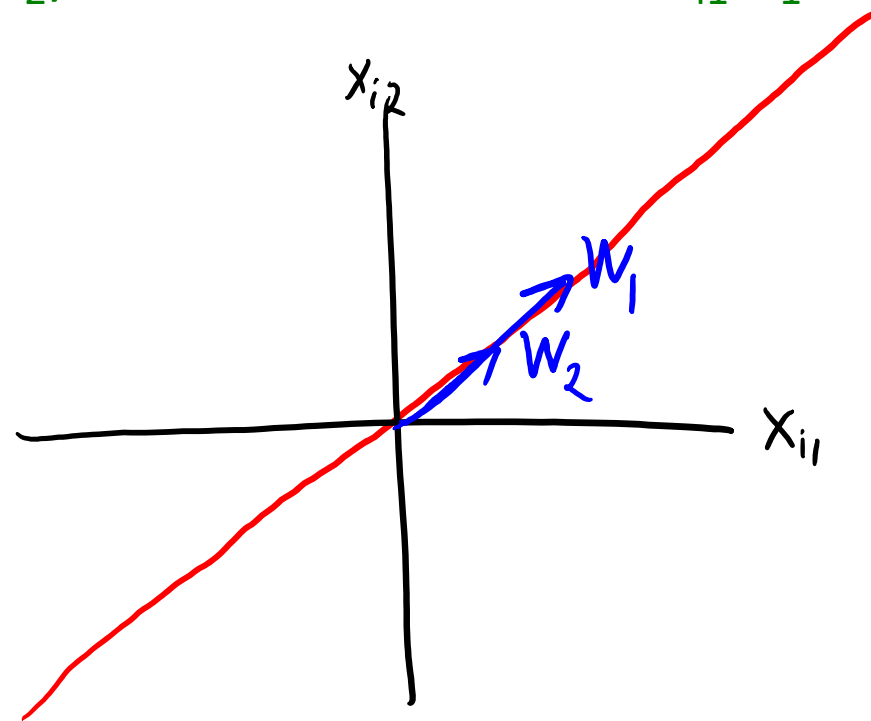
- Consider two vector w_1 and w_2 ($k=2$).
 - The $\text{span}(w_1, w_2)$ is all vectors of form $z_{i1}w_1 + z_{i2}w_2$ for a scalars z_{i1} and z_{i2} .



- For most non-zero 2d vectors, $\text{span}(w_1, w_2)$ is a plane.
 - In the case of two vectors in $\mathbb{R}^2$, the plane will be *all* of $\mathbb{R}^2$.

Span of 2 Vectors

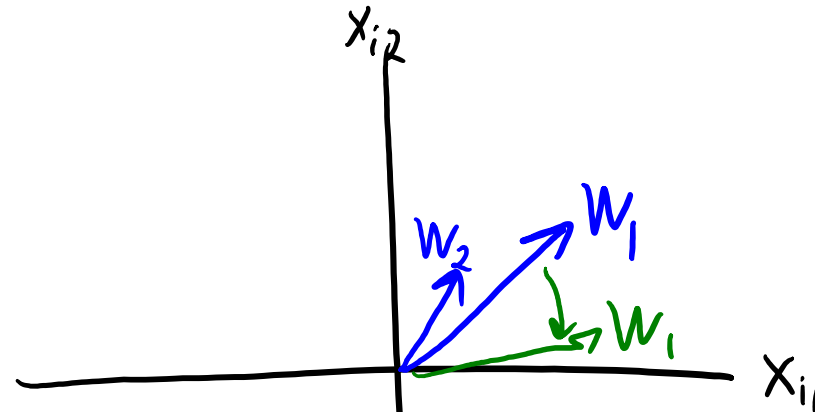
- Consider two vector w_1 and w_2 ($k=2$).
 - The $\text{span}(w_1, w_2)$ is all vectors of form $z_{i1}w_1 + z_{i2}w_2$ for a scalars z_{i1} and z_{i2} .



- For most non-zero 2d vectors, $\text{span}(w_1, w_2)$ is plane.
 - Exception is if w_2 is in span of w_1 ("collinear"), then $\text{span}(w_1, w_2)$ is just a line.

Span of 2 Vectors

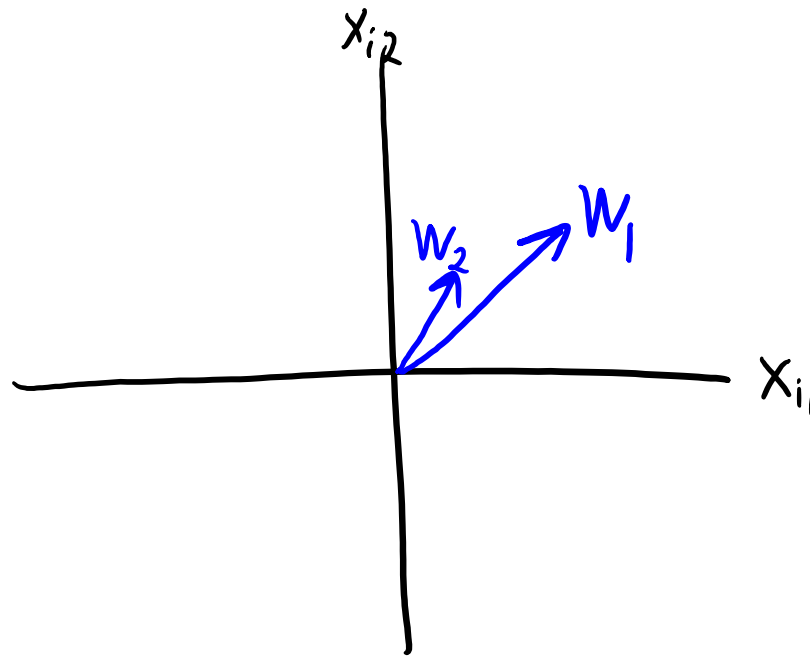
- Consider two vector w_1 and w_2 ($k=2$).
 - The $\text{span}(w_1, w_2)$ is all vectors of form $z_{i1}w_1 + z_{i2}w_2$ for a scalars z_{i1} and z_{i2} .



- New issues for PCA ($k \geq 2$):
 - We have **label switching**: $\text{span}(w_1, w_2) = \text{span}(w_2, w_1)$.
 - We can **rotate factors** within the plane (if not rotated to be collinear).

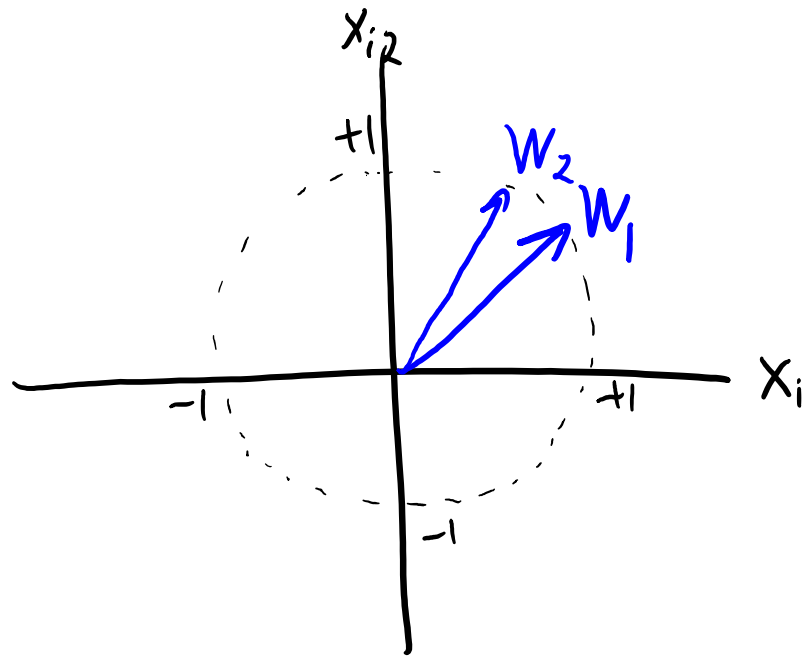
Span of 2 Vectors

- 2 tricks to make vectors defining a plane “more unique”:
 - Normalization: enforce that $||w_1|| = 1$ and $||w_2|| = 1$.



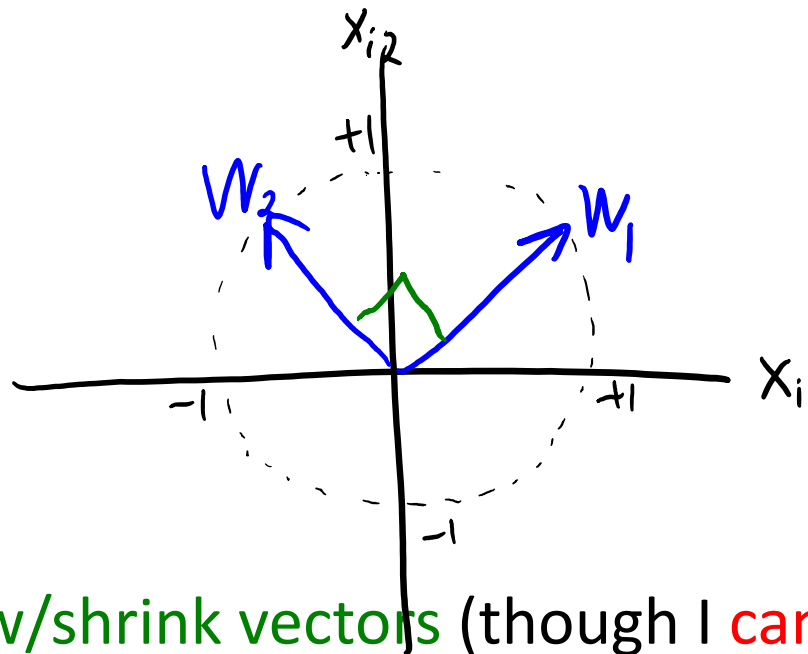
Span of 2 Vectors

- 2 tricks to make vectors defining a plane “more unique”:
 - **Normalization**: enforce that $||w_1|| = 1$ and $||w_2|| = 1$.



Span of 2 Vectors

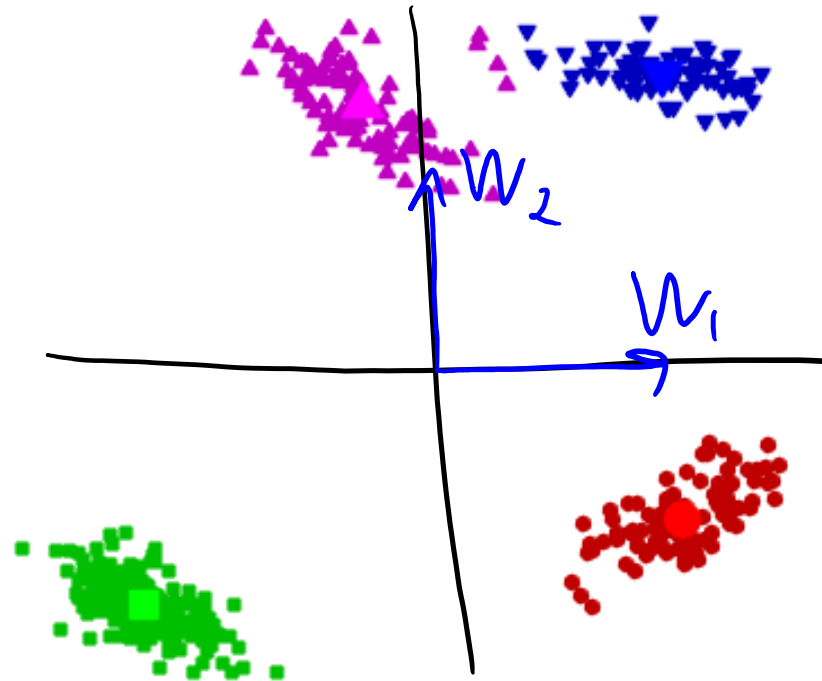
- 2 tricks to make vectors defining a plane “more unique”:
 - **Normalization**: enforce that $||w_1|| = 1$ and $||w_2|| = 1$.
 - **Orthogonality**: enforce that $w_1^T w_2 = 0$ (“perpendicular”).



- Now I **can't grow/shrink vectors** (though I **can still reflect**).
- Now I **can't rotate one vector** (but I **can still rotate *both***).

Digression: PCA only makes sense for $k \leq d$

- Remember our clustering dataset with 4 clusters:



- It **doesn't make sense to use PCA with $k=4$** on this dataset.
 - We **only need two vectors** $[1\ 0]$ and $[0\ 1]$ to exactly represent all 2d points.
 - With $k=2$, I could set $Z=X$ and $W=I$ to get $X=ZW$ exactly.

Span in Higher Dimensions

- In higher-dimensional spaces:
 - Span of 1 non-zero vector w_1 is a line.
 - Span of 2 non-zero vectors w_1 and w_2 is a plane (if not collinear).
 - Can be visualized as a 2D plot.
 - Span of 3 non-zero vectors $\{w_1, w_2, w_3\}$ is a 3d space (if not “coplanar”).
 - ...
- This is how the W matrix in PCA defines lines, planes, spaces, etc.
 - Each time we increase ‘k’, we add an extra “dimension” to the “subspace”.

Summary

- PCA geometry:
 - With $k=1$, PCA projects data onto a line .
 - With $k=2$, PCA projects data onto a plane.
- PCA objective:
 - Minimizes squared error between elements of X and elements of ZW .
- Choosing 'k':
 - We can choose 'k' to explain “percentage of variance” in the data.
- PCA non-uniqueness:
 - Due to scaling, rotation, and label switching.
- Next time: cancer signatures and NBA shot charts.

Making PCA Unique

- PCA implementations add **constraints to make solution unique**:
 - **Normalization**: we enforce that $\|w_c\| = 1$.
 - **Orthogonality**: we enforce that $w_c^T w_{c'} = 0$ for all $c \neq c'$.
 - **Sequential fitting**: We **first fit w_1** (“first principal component”) giving a line.
 - **Then fit w_2 given w_1** (“second principal component”) giving a plane.
 - **Then we fit w_3 given w_1 and w_2** (“third principal component”) giving a space.
 - ...
- Even with all this, the solution is **only unique up to sign changes**:
 - I can still replace any w_c by $-w_c$:
 - $-w_c$ is normalized, is orthogonal to the other $w_{c'}$, and spans the same space.
 - Possible fix: **require that first non-zero element of each w_c is positive**.
 - And this is assuming you don't have repeated singular values.
 - In that case you can rotate the repeated ones within the same plane.